

Ratio and Proportion: Scale factors and Similar Shapes

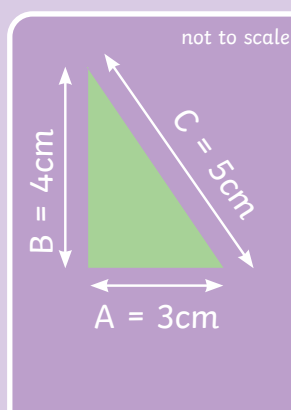
What is a scale factor?

The word **scale** refers to the relative size of a shape. You might notice drawings or maps that say 'not to scale'. The **scale factor** is the term used to describe the ratio between corresponding side lengths in similar shapes.

Using the scale factor can help us to identify similar shapes and scale shapes up (enlarge) or scale shapes down (reduce).

If we use a scale factor bigger than 1, the new shape will be **bigger**.

If we use a scale factor smaller than 1, the new shape will be **smaller**.



Let's take a look at this shape. The sides are labelled A, B and C.

Side A = 3cm

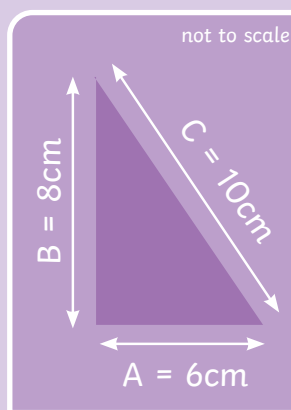
Side B = 4cm

Side C = 5cm

We can write this as a ratio:

A:B:C

3:4:5



Here's another shape. The sides are also labelled A, B and C.

Side A = 6cm

Side B = 8cm

Side C = 10cm

We can write this as a ratio too:

A:B:C

6:8:10

We hope you find the information on our website and resources useful. As far as possible, the contents of this resource are reflective of current professional research. However, please be aware that every child is different and information can quickly become out of date. The information given here is intended for general guidance purposes only and may not apply to your specific situation.

Ratio and Proportion: Scale Factors and Similar Shapes

$$\begin{array}{c} A : B : C \\ 6 : 8 : 10 \\ \div 2 \\ \hline 3 : 4 : 5 \end{array}$$

Using our understanding of ratio, we can simplify the ratio of the second shape from 6:8:10 to 3:4:8. That's the same ratio as the first shape!

The lengths of the corresponding sides in both shapes are in the same ratio, which means that the corresponding angles are the same too. This means that the green triangle and the purple triangle are **similar shapes**.

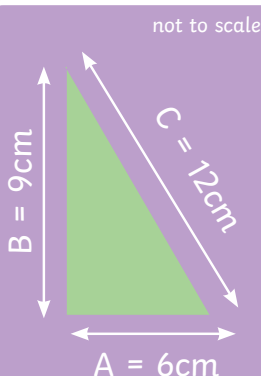
$$\begin{array}{ccc} \text{Side A} & \text{Side B} & \text{Side C} \\ 3\text{cm} & 4\text{cm} & 5\text{cm} \\ \downarrow & \downarrow & \downarrow \\ \hline & \times 2 & \\ \downarrow & \downarrow & \downarrow \\ 6\text{cm} & 8\text{cm} & 10\text{cm} \end{array}$$

The purple triangle is an enlarged version of the green triangle.

The scale factor is the number which we multiplied each of the corresponding side lengths by in order to create the enlarged shape.

In this case, we multiplied each side by 2. This means that the purple triangle is two times as big as the green one.

The scale factor of enlargement is 2.



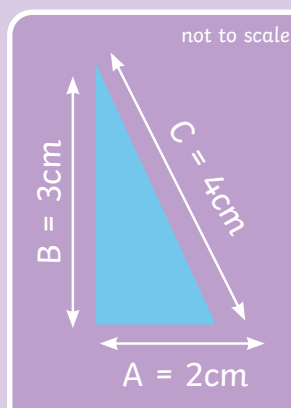
When we want to reduce the size of a shape, the scale factor will be less than 1. This shape has sides labelled A, B and C.

Side A = 6cm
Side B = 9cm
Side C = 12cm

We can write this as a ratio:

A:B:C
6:9:12

Ratio and Proportion: Scale factors and Similar Shapes



Here's another shape. The sides are also labelled A, B and C.

Side A = 2cm

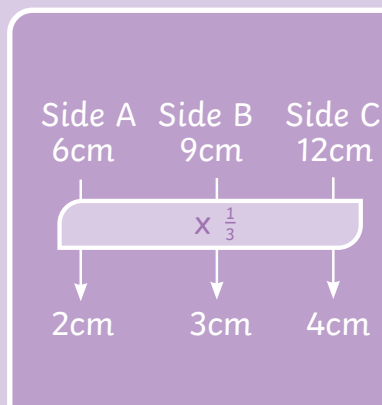
Side B = 3cm

Side C = 4cm

We can write this as a ratio too:

A:B:C

2:3:4



The blue triangle is a reduced version of the green triangle.

The scale factor is the number which we multiplied each of the corresponding side lengths by in order to create the new shape.

In this case, we multiplied each side by $\frac{1}{3}$. Multiplying by $\frac{1}{3}$ is the same as dividing by 3. This means that the blue triangle is three times smaller than the green one.

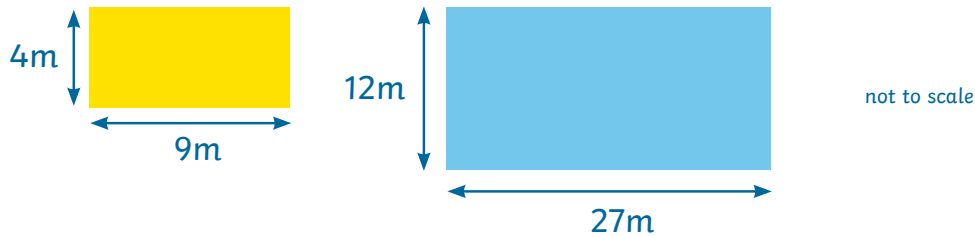
The scale factor from the green to the blue triangle is $\frac{1}{3}$.

Fractions, Decimals, Percentages: Simplifying Fractions

Questions

Use your knowledge of scale factor and similar shapes to solve these problems.

What scale factor has the yellow rectangle been enlarged by to make the blue rectangle?

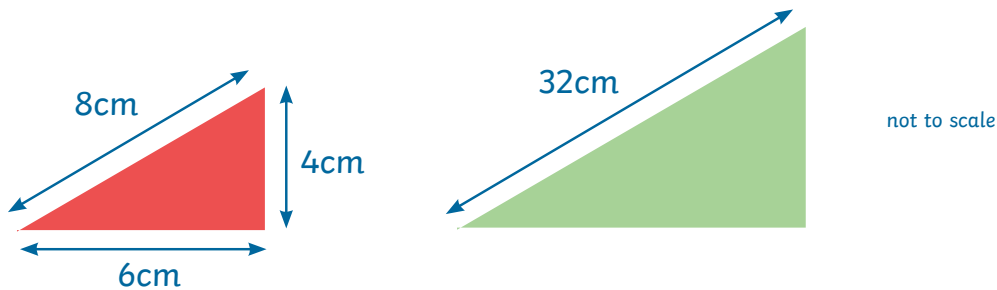


The ratio of the sides of the yellow rectangle is _____.

The unsimplified ratio of the sides of the blue rectangle is _____.

The blue rectangle is _____ times as big, so the scale factor of enlargement is _____.

What scale factor has the red triangle been enlarged by to make the green triangle?



The unsimplified ratio of the sides of the red triangle is _____.

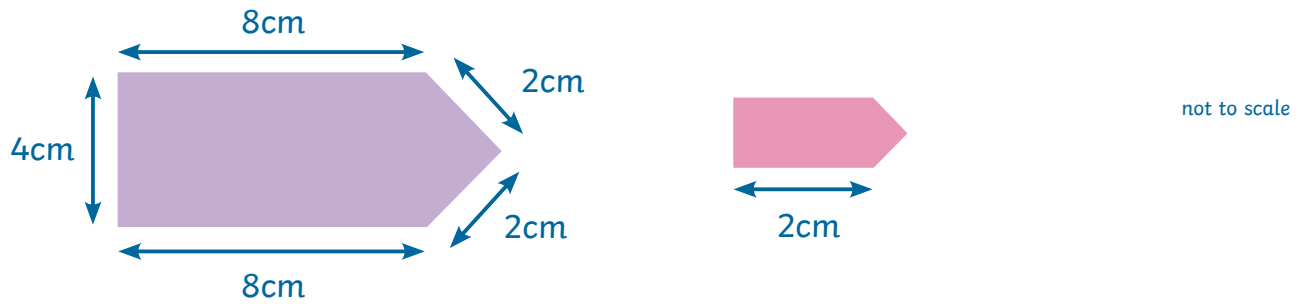
The unsimplified ratio of the sides of the green triangle is _____.

The simplified ratio of both triangles is _____.

The green triangle is _____ times as big, so the scale factor of enlargement is _____.

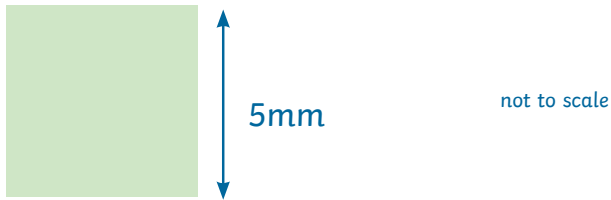
Fractions, Decimals, Percentages: Simplifying fractions

What scale factor has the purple shape been reduced by to make the pink shape?



The pink shape is _____ the size of the purple shape, so the scale factor is _____.

Look at this shape.



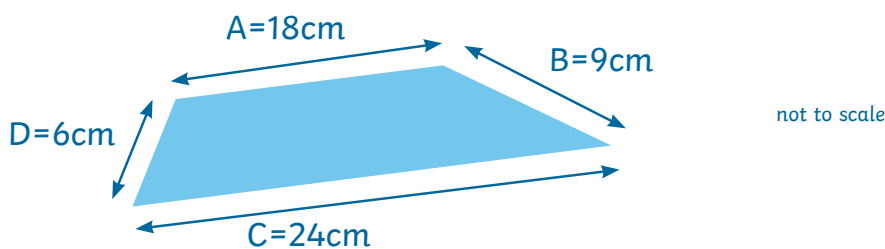
Write down the length the sides of the new shape would be if this shape was enlarged by:

A scale factor of 5: _____ mm

A scale factor of 10: _____ mm

A scale factor of 12: _____ mm

Look at this shape.

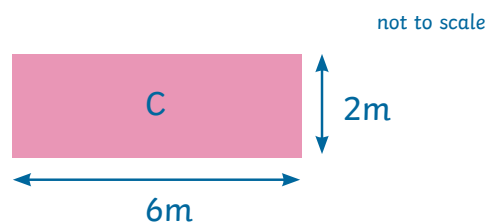
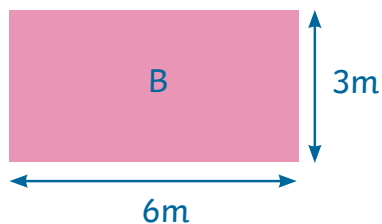
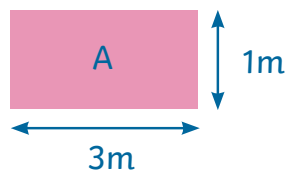


Write down the lengths of the sides of some new shapes if they have been drawn with:

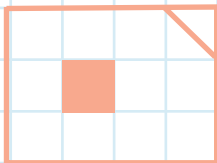
a scale factor of $\frac{1}{2}$:	a scale factor of $\frac{1}{3}$:	a scale factor of $\frac{1}{10}$:
A = _____ cm	A = _____ cm	A = _____ cm
B = _____ cm	B = _____ cm	B = _____ cm
C = _____ cm	C = _____ cm	C = _____ cm
D = _____ cm	D = _____ cm	D = _____ cm

Fractions, Decimals, Percentages: Simplifying Fractions

Look at these three shapes. Are they similar shapes?



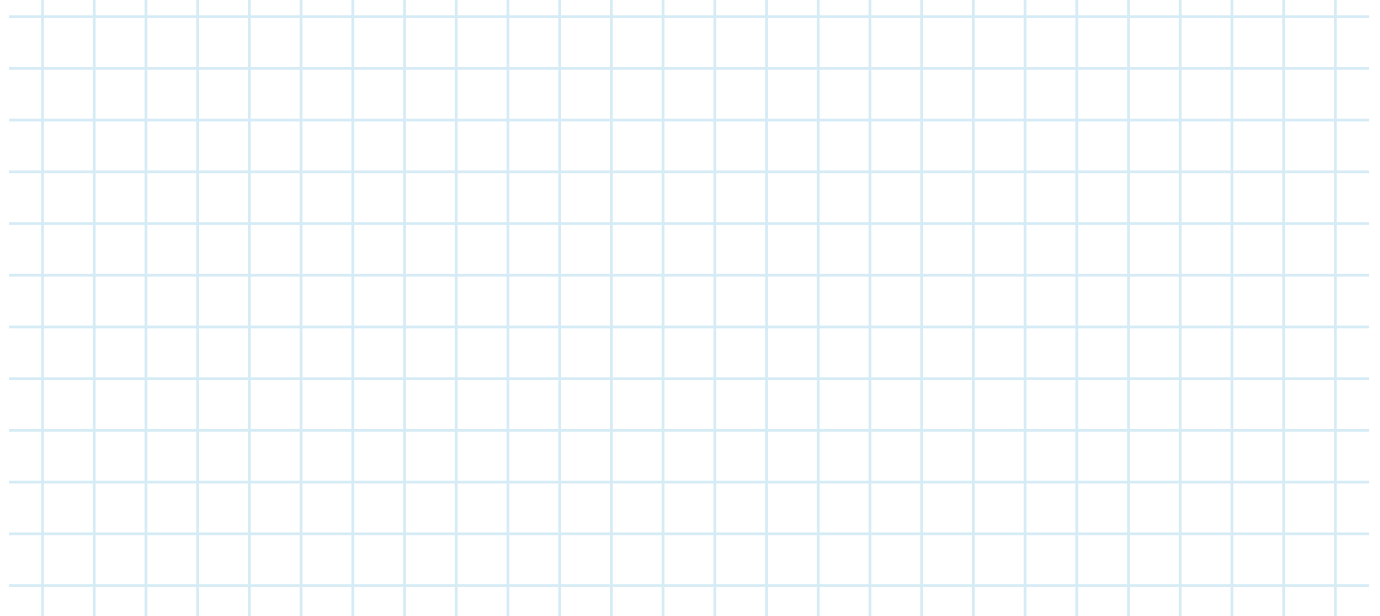
Enlarge this shape by a scale factor of 2.



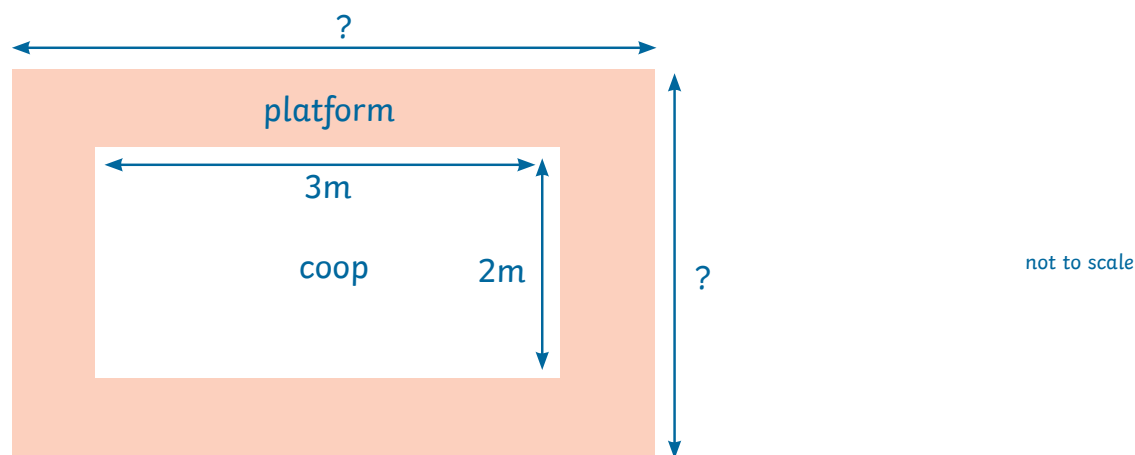
Fractions, Decimals, Percentages: Simplifying fractions

I am thinking of a shape. It has four equal sides of 4cm. I have enlarged my shape by a scale factor of 2.5.

Draw my shape.

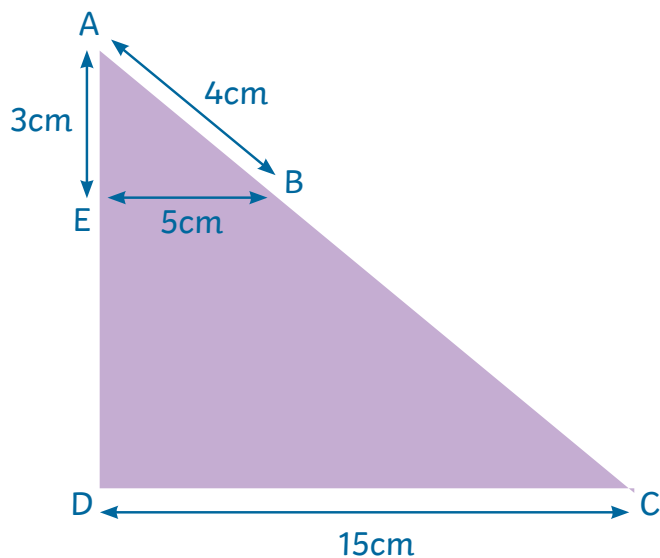


This is a drawing of a chicken coop. The coop sits on a raised platform which is a similar shape as the coop but is enlarged by a scale factor of 1.5. What is the perimeter of the platform?



Fractions, Decimals, Percentages: Simplifying Fractions

This diagram shows two similar shapes - triangle ABE and triangle ACD.



not to scale

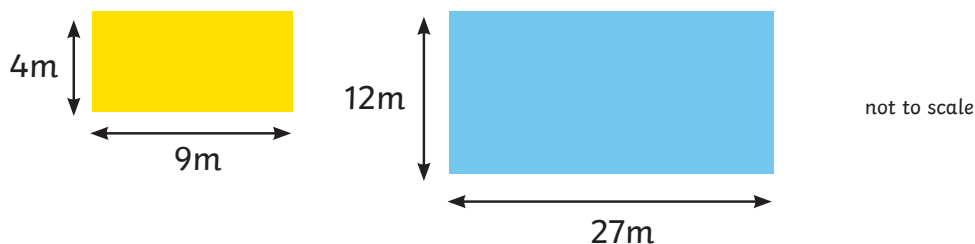
Use the scale factor to find out the length of sides AC and AD.

AC = _____ cm

AD = _____ cm

Ratio and Proportion: Scale Factors and Similar Shapes

What scale factor has the yellow rectangle been enlarged by to make the blue rectangle?

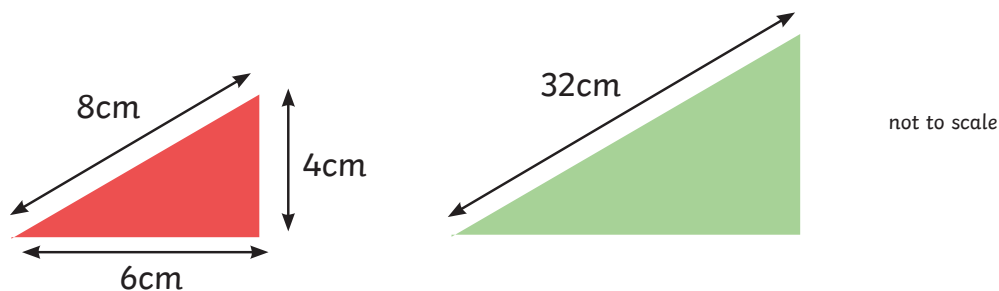


The ratio of the sides of the yellow rectangle is **4:9:4:9**.

The unsimplified ratio of the sides of the blue rectangle is **12:27:12:27**.

The blue rectangle is **3** times as big, so the scale factor of enlargement is **3**.

What scale factor has the red triangle been enlarged by to make the green triangle?



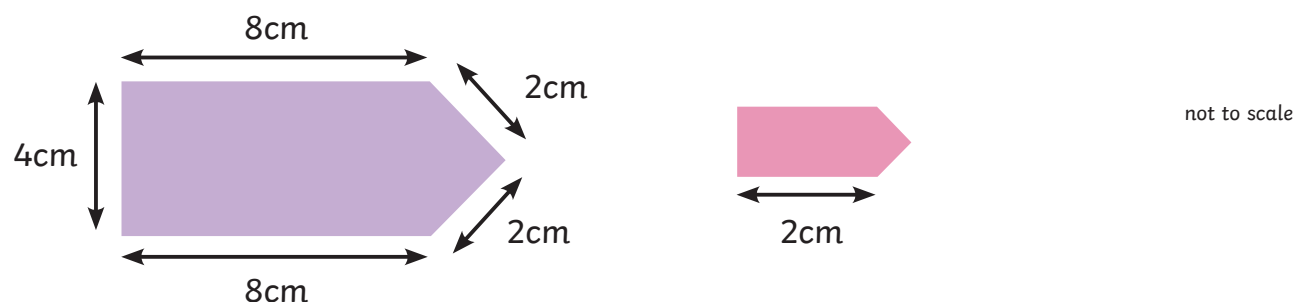
The unsimplified ratio of the sides of the red triangle is **8:4:6**.

The unsimplified ratio of the sides of the green triangle is **32:16:24**.

The simplified ratio of both triangles is **4:2:3**.

The green triangle is **4** times as big, so the scale factor of enlargement is **4**.

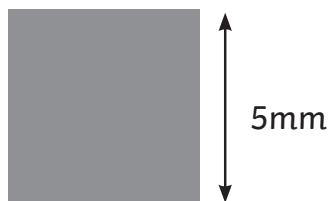
What scale factor has the purple shape been reduced by to make the pink shape?



The pink shape is $\frac{1}{4}$ the size of the purple shape, so the scale factor is $\frac{1}{4}$.

Ratio and Proportion: Scale Factors and Similar Shapes

Look at this shape.



not to scale

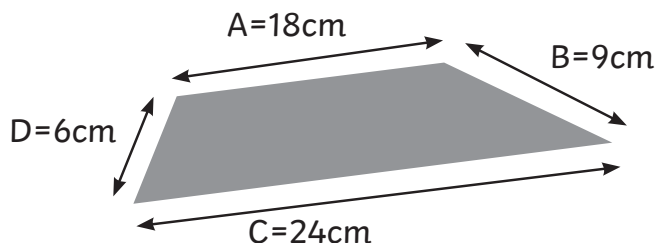
Write down the length the sides of the new shape would be if this shape was enlarged by:

A scale factor of 5: **25** mm

A scale factor of 10: **50** mm

A scale factor of 12: **60** mm

Look at this shape.



not to scale

Write down the lengths of the sides of some new shapes if they have been drawn with:

a scale factor of $\frac{1}{2}$:

A = **9** cm

B = **4.5** cm

C = **12** cm

D = **3** cm

a scale factor of $\frac{1}{3}$:

A = **6** cm

B = **3** cm

C = **8** cm

D = **2** cm

a scale factor of $\frac{1}{10}$:

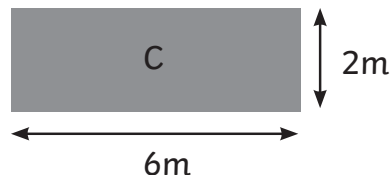
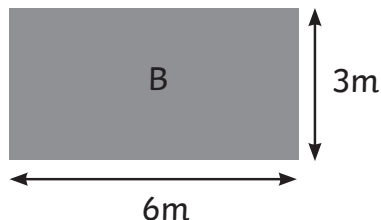
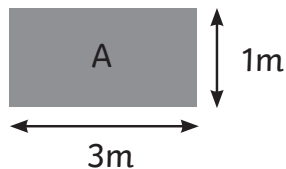
A = **1.8** cm

B = **0.9** cm

C = **2.4** cm

D = **0.6** cm

Look at these three shapes. Are they similar shapes?

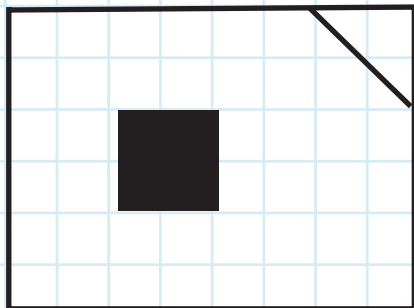
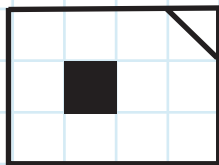


not to scale

Shapes A and C are similar shapes - shape C is shape A enlarged by a scale factor of 2.
Shape B is not similar to A and C.

Ratio and Proportion: Scale Factors and Similar Shapes

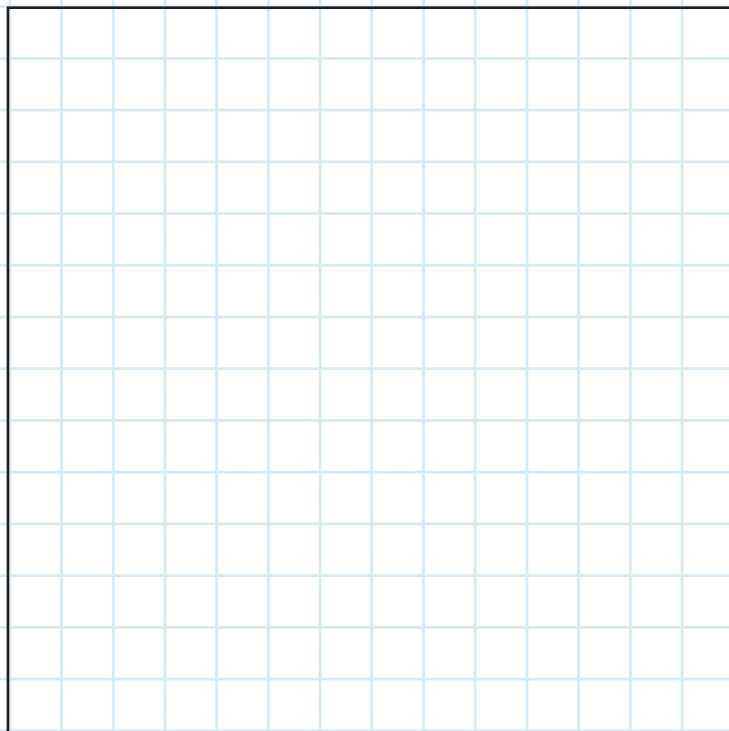
Enlarge this shape by a scale factor of 2.



I am thinking of a shape. It has four equal sides of 4cm. I have enlarged my shape by a scale factor of 2.5.

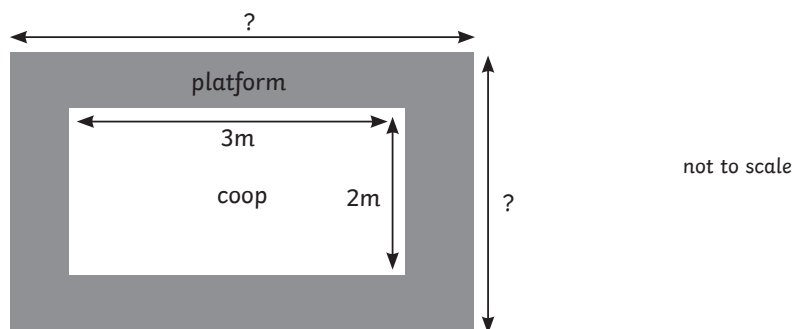
Draw my shape.

As long as your shape has four equal sides of 10cm, it can be in any orientation.



Ratio and Proportion: Scale Factors and Similar Shapes

This is a drawing of a chicken coop. The coop sits on a raised platform which is a similar shape as the coop but is enlarged by a scale factor of 1.5. What is the perimeter of the platform?



The size of the coop has been enlarged by a scale factor of 1.5 to create the size of the platform.

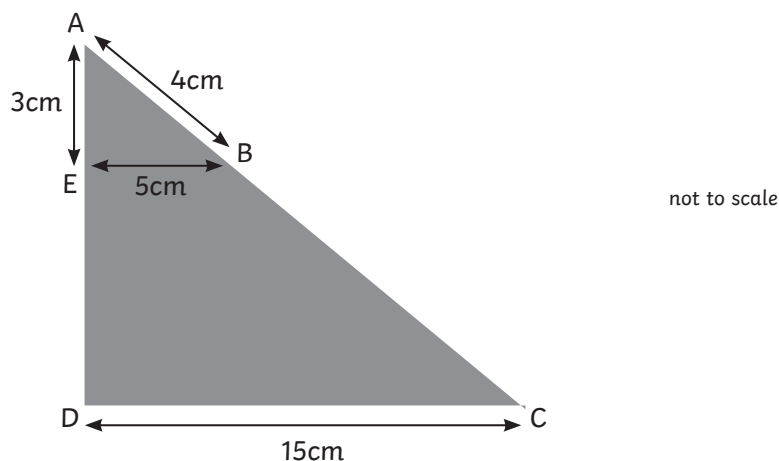
$$3\text{m} \times 1.5 = 4.5\text{m}$$

$$2\text{m} \times 1.5 = 3\text{m}$$

The long sides of the platform are 4.5m long and the short sides are 3m long.

$$4.5\text{m} + 4.5\text{m} + 3\text{m} + 3\text{m} = 15\text{m}$$

This diagram shows two similar shapes - triangle ABE and triangle ACD.



Use the scale factor to find out the length of sides AC and AD.

Side EB = 5cm and the corresponding side DC = 15cm.

$5\text{cm} \times 3 = 15\text{cm}$ so the scale factor is 3.

AB is 4cm so the corresponding side AC must be $4\text{cm} \times 3$.

AE = 3cm so the corresponding side AD must be $3\text{cm} \times 3$.

$$AC = 12\text{cm}$$

$$AD = 9\text{cm}$$